MIL-M-82433A(OS)
5 April 1972
Superseding
MIL-M-82433(OS)
10 May 1966

MILITARY SPECIFICATION

MOTOR, DIRECT CURRENT

(For Torpedo Mk 46 Mod 1)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers requirements for the procurement of one type of armature-controlled dc motor (see 6.1.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, Gen Spec for

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-440	Soldering Techniques for Standard Type Solder Terminals

FSC 1355

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

MIL-STD-889	Dissimilar Metals
MS-16997	Screw, Cap, Socket Head, Hexagon, Alloy Steel, Cadmium plated UNC-3A
MS-21042-08	Nut, Self-Locking, 450° F, Reduced Hexagon, Reduced Height, Ring Base, Noncorrosion Resistant Steel

Federal

Fed. Test Method STD-151	Metals; Test Methods

DRAWINGS

Naval Ordnance Systems Command

NAVORD Dwg 2540053	Motor, Direct Current, Specification Control Drawing

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of the specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Aircraft Standards

NAS-620	Washer, Plain

(National Aircraft Standards maybe obtained from National Standards Association, Inc., 1321 Fourteenth St. N.W., Washington,D.C. 20005.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample shall be provided and shall

(a) Be manufactured according to procedures proposed for the production lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

(b) Consist of nine motors.

(c) Be tested as specified in Section 4 herein and is for the purpose of determining, prior to starting production, that the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2 Basic requirements

3.2.1 Contractor-furnished data. No data is required by this specification or by applicable documents referenced in Section 2 unless specified in the contract or order (see 6.2).

3.2.2 Materials and treatments of materials and parts. Unless otherwise specified in the contract or order, all materials and treatments of materiels and parts used in the manufacture of the motor shall be in accordance with the requirements specified herein and with the requirements and documents shown on the Specification Control Drawing 2540053.

3.2.3 Inspection and test equipment and facilities. The supplier shall furnish and maintain all necessary inspection and test equipment facilities, and shall provide personnel for performing all quality conformance tests specified herein. The test equipment shall be adequate to permit performance of the required quality conformance tests for the motor.

3.2.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in contact with each other. Dissimilar metals are defined in MIL-STD-89.

3.2.5 Soldering. All solder connections shall be made in accordance with MIL-STD-440.

3.2.6 Fungus proof material. Materials that are nutrients for fungi shall not be used unless necessary. Where used and not hermetically sealed, they shall be treated with a fungicidal agent.

3.2.7 Fuel comparability. The exterior of the motor may be exposed to splashing by the torpedo propellant, a liquid fuel known as Otto Fuel Composition II. This is a nitrate ester based propellant diluted with straightchain hydrocarbons. The fuel is compatible with all metals and with both silicone and fluorc carbon-based lubricants. The fuel is compatible with some nonmetals and incompatible, more or less, with others. The approved nonmetallic materials are listed below.

(a) Fluorc carbons (Teflon and Kel-F)

(b) Polyethylene

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

(c) Neoprene

(d) Glass

In the event that other nonmetals are proposed for use in the exterior structure of the motor, samples of such materials shall be submitted to the procuring authority for test and approval.

3.3 Performance and product requirements

3.3.1 Function. The motor drives a control surface through a worm gear actuator in response to the control voltage which reverses in polarity at approximately 5 Hertz (Hz). The motor is expected to have an integral gear reduction unit to obtain the required speed and torque.

3.3.2 Electrical requirements

3.3.2.1 Supply voltage. The specification center supply or input voltage shall-be 38 volts. The supply voltage range shall be 351 to 41 volts dc.

3.3.2.2 No-load speed. The output shaft speed at no-load condition and 38 volts input shall be 120 ± 5 rpm, taking the average of two directions. The difference of speeds in the two directions shall not exceed 4 rpm. The average of speeds with 19 volts input at no-load shall be within 50 f5 percent of the average of speeds obtained with 38 volts input. The sampling period shall not exceed 3 seconds. The motor shall be at room temperature (65-95 degrees Fahrenheit (F)).

3.5.2.3 No-load current. The motor current at no-load condition, 38 volts input, and at room temperature shall be less than 0.220 ampere. Motor currents, in two directions, shall hae an average of less than 0.220 ampere and the difference of the two currents shall be less that 15 percent of the average value.

3.3.2.4 Loading. The rated load shall be 100 ounce-inches. The normal loading is expected to vary from 0 to 100 ounce-inches.

3.3.2.5 Speed

3.3.2.5.1 Speed at rated load. The output shaft speed in either direction at rated load of 100 ounce-inches, with 38 volts input and with motor initially at room temperature (65-95 degrees F), shall be greater than 83 percent of the average no-load speed (3.3.2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

3.3.2.5.2 Speed at overload. The output saft speed in either direction at an overload of 300 f6 inch-ounces, with 38 volts applied and with the motor initially at room temperature (65-95 degrees F), shall be greater than 50 percent of the average no-load speed (3.3.2.2).

3.3.2.6 Stall current. The design stall current at room temperature with 38 volts input shall not exceed 3.0 amperes.

3.3.2.7 Motor Stability. The motor shall be capable of repeated sudden reversals of direction of rotation (“plugging”) at maximum input voltage (41 vdc) without deterioration of the permanent magnet field. This shall be indicated by stability of the no-load speed of the motor, before and after a series of instantaneous reversals of polarity of the input voltage.

3.3.2.8 Motor Time Constant. Under no-load condition, at room temperature, and with 38 volts step input, the motor time constant for either direction of rotation shall not exceed 12 milliseconds. The time constant is defined as the time required for the motor to accelerate from rest to 63 percent of final speed after application of a step voltage input.

3.3.2.9 Shaft rotation and terminal marking. One input terminal or lead shall be marked (+) and/or color coded red in such a way that with positive voltage applied at this connection the output shaft rotation shall be in the counterclockwise direction when looking at the shaft end of the motor.

3.3.2.10 Insulation resistance. The resistance between the motor circuit and the motor frame shall exceed 100 megohms when tested at 200 vdc potential.

3.3.3 Duty cycle. The duty cycle shall consist of 10 minutes operating time followed by a minimum of 30 minutes nonoperating time. The following conditions shall exist during the operating time: the motor is energized by a reversing waveform approximating a square wave at 5.00 ±0.25 Hertz (Hz) at 38 + 0-2.0 volts line-to-line. The waveform shall have a reversing time of less than 100 microseconds. Load applied to the output shaft shall increase linearly with angular rotation in each direction from the null position at a rate of 2.0 ±0.2 ounce-inches per degree. The null position is defined as the point of zero load, or where the load changes direction. The ambient temperature may be at any temperature within the operating range O to ±200 degrees F.

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:07Z
Check the source to verify that this is the current version before use.

3.3.4 Operating Life. The motor operating life shall be 25 hours of cumulative operating time through repeated duty cycles, as specified in 3.3.3, for a total of 150 duty cycles. Acceptable performance of operating life is defined: the motor shall satisfactorily complete the 25 hours of operation and, subsequently, the quality conformance tests of 4.7 with the following modifications:

(a) Due to expected buildup of brush residue during motor operation, the requirement on insulation resistance (4.7.3.1) shall be not less than 10,000 ohms for motors that have been subjected to life test.

(b) The requirement on no-load speed (4.7.3.2) for motors that have been subjected to life test shall be 127 rpm maximum and 113 rpm minimum. The speeds in opposite directions, after life test, shall not differ,by more than 6 rpm.

3.3.5 Strength of wire or terminal connections. Each of the electrical connecting wires or terminals shall be capable of withstanding a pull test of five pounds.

3.3.6 Brush fitment. Prior to quality conformance testing, in order to fit the brushes to the commutator and stabilize motor characteristics, each motor shall be opera-ted at 26.0 ±4.0 vdc for from 10 to 30 minutes in each direction, repeated for a number of such cycles to give a total run-in time of at least 12 hours.

3.3.7 Commutation noise. The ac component of the motor current, when running at no-load and at nominal voltage, shall contain a minimum of radiofrequency components and shall be relatively free of other sharp discontinuities. The commutation noise-level shall not exceed 0.75 volt when measured as directed in 4.7.3.10.

3.4 Environmental requirements. The motor shall be capable of meeting all performance and product requirements of this specification after being exposed to the following environments and conditions:

NOTE

Motor shall be attached to fixture by its normal mounting features: to mount motor use 4 each, MS-16997-33 Scr cap Soc Hd; MS-21042-08 Nut-Self Lock; and NAS 620-8 Washer, Plain. Torque the mounting hardware to 25 ±5 inch pounds. (See Figure 1.)

3.4.1 Vibration

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:07Z
Check the source to verify that this is the current version before use.

3.4.1.1 Vibration; ready and ready nonoperating condition. The motor, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the vibrations defined in Table I. The motor shall be vibrated for 180 minutes each in its longitudinal axis and in two mutually perpendicular transverse axes at any temperature within the range of -45°F to +140 F. Vibration is defined in 6.3.1.

3.4.1.2 Vibration; packed. The motor shall be capable of meeting the performance requirements after being subjected, in the packed condition (see 5.2), to the vibrations shown in Table I.

Table I

Vibration Schedule

Condition (see 6.3.2)	Double Amplitude (in.) Minimum	Acceleration (g's) Minimum	Frequency Range (Hz)
Packed	---	1.0	5 minimum
Ready Nonoperating and Ready	0.2	---	5 - 17
	---	3.0	17 - 40
	0.036	---	40- 74
	---	10	74 - 500

3.4.2 Shock. The packed motor (see 5.2) shall be capable of withstanding the shock of being dropped on a nonyielding surface from a height of 36 inches. Additionally, the motor, in the condition shown in Table II. shall withstand shocks specified in Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

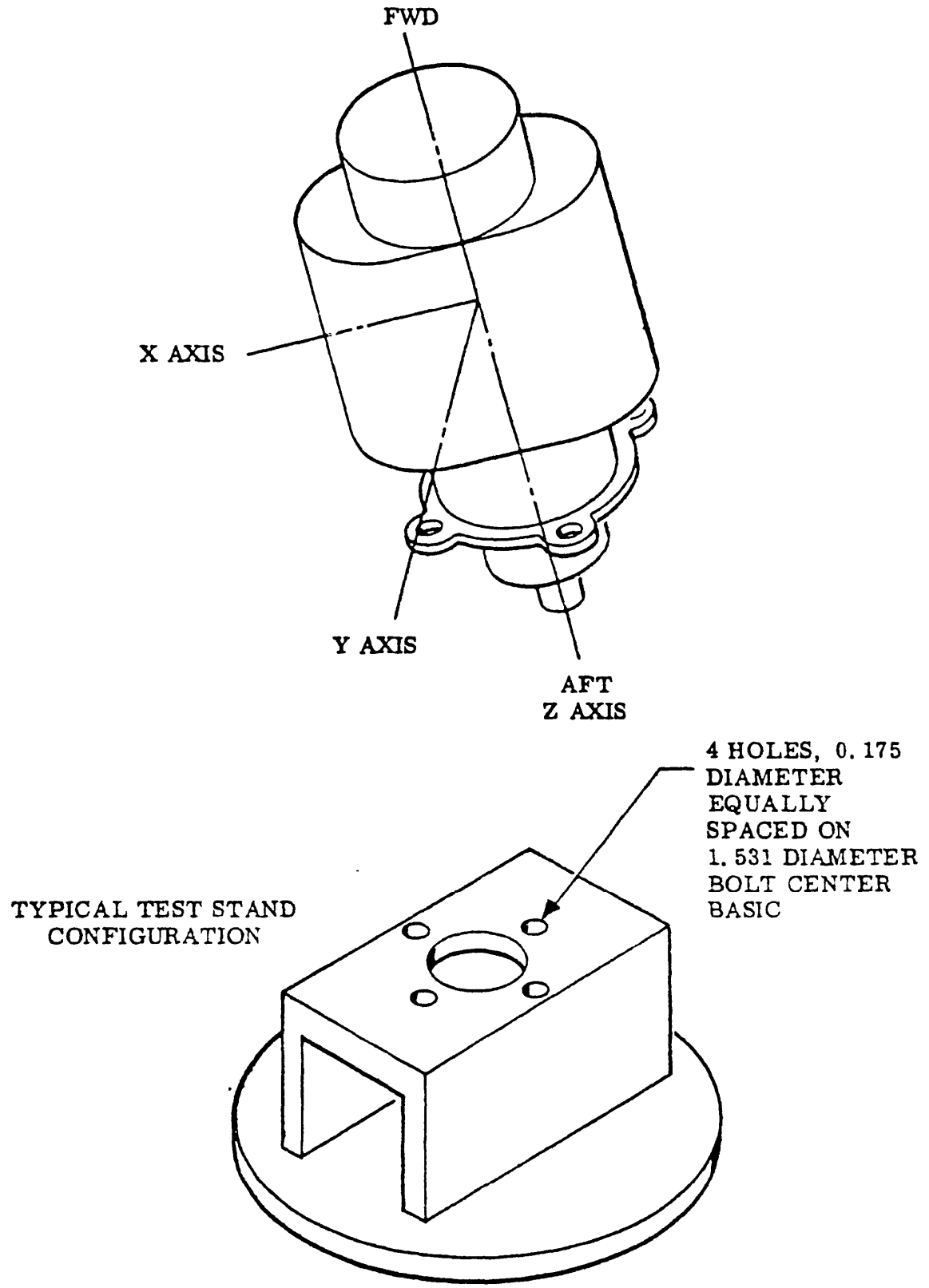


Figure 1. Motor Axis Orientation and Mounting Details

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

Table II

Shock Schedule

Condition (see 6.3.2)	Peak Magnitude (g's)	Minimum Duration ms	Acceleration Direction*	Number of Impact
Ready nonoperating	60 ±10%	8.0 to 10.0	X,Y or Z Axis	9
Ready	45 ±10%	18.0 to 20.0	Z Axis-Forward	3
	85 ±10%	0.5 to 1.5	X, Y or Z Axi 6	6
	300 ±10%	0.5 to 1.5	Z Axis-Aft	3

*See Figure 1 for axes of' motor.

3.4.3 Acceleration. The motor shall withstand the accelerations specified in Table 111 (see Figure 1).

Table III

Acceleration Schedule

Condition see 6.3.2)	Magnitude (g's)	Acceleration Direction
Ready	25 ±10%	Z Axis - Forward
	35 ±10%	Z Axis -Aft

3.4.4 Temperature. The motor shall be capable of withstanding the temperatures for the following specified conditions:

3.4.4.1 Nonoperating and ready nonoperating. The motor shall be capable of meeting the performance requirements subsequent to withstanding, in the nonoperating condition, environmental temperatures between -45°F and +140°F. Temperature is defined in 6.3.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

3.4. 4.2 Operating. The motor shall be capable of operating as specified in 3.3.3 at ambient temperatures from 0 degrees to +200 degrees F.

3.4.5 Humidity. The motor shall withstand exposure to 95 percent relative humidity at 110 degrees F for 8 hours without degradation of performance.

3.4.6 Enclosure. The motor shall have a splashproof enclosure that is sufficiently watertight to withstand the test of 4.5.2.6.

3.4.7 Pressure. The motor shall be capable of withstanding pressures in the ready, nonoperating condition in the range of 2.7 psia to 20 psia; the maximum rate of change of pressure shall be 0.5 psi per second. Pressures in the operating condition will be in the range 15 to 20 psia.

3.5 Workmanship. Workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with requirements of this specification. All parts used in the motor shall be thoroughly clean and free of excess material, chips, and loose or spattered foreign materials. Standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections of the motor shall be classified as follows:

(a) Reproduction inspection

(1) Preproduction sample (see 4.3.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

(2) Reproduction tests (see 4.5)

(b) Periodic production inspection

(1) Periodic production sampling (see 4.3.2)

(2) Periodic production tests (see 4.6)

(c) Quality conformance inspection

(1) Quality conformance sampling (see 4.3.3)

(2) Quality conformance tests and examinations (see 4.7)

4.3 Sampling

4.3.1 Preproduction sample. The preproduction sample of nine motors furnished in accordance with 3.1 shall be subjected to the preproduction tests detailed in 4.5 at an activity acceptable to the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the motor by supplier, prior to approval of the preproduction sample, shall be at supplier's risk.

4.3.2 Periodic production sampling. Periodic production samples shall be selected from units having satisfactorily passed quality conformance tests and examinations of 4.7. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

(a) Step 1. Upon initiation of production, randomly select one unit from each group of 30 units. Upon satisfactory compliance of samples selected from three consecutive groups, shift to Step 2. Each sample so selected shall be numbered consecutively starting with Sample Number 1.

(b) Step 2. Randomly select one unit from each group of 70 units. Continue with Step 2 for the balance of the contract or until a defective unit is found. Each sample so selected shall be numbered consecutively starting with a sample number one greater than the highest sample number selected in Step 1. When a defective unit (see 6.3.8) us fiynd and inspection is resumed, return to Step 1 and proceed as indicated.

4.3.3 Quality conformance sampling. The tests and examinations specified in 4.7 shall be conducted on each motor to be delivered; therefore, there - shall be no sampling plan for acceptance. Defective units shall be rejected

4.4 Test conditions. Unless otherwise specified, the motor shall be subjected to tests under following conditions:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

(a) Temperature	Room ambient (65 to 95°F)
(b) Humidity (relative)	Room ambient, up to 95 percent
(c) Altitude	Normal ground atmospheric pressure
(d) Vibrations	None

4.5 Reproduction tests. The motors within the preproduction sample shall be identified as motors 1, 2, 3, 4, 5, etc., and shall be subjected to respective tests as specified in Table IV. Certified inspection data shall consist of following:

(a) Date of tests and names of' persons conducting the tests.

(b) Run or test plan used

(c) Serial numbers of the motors.

(d) Test results of the preproduction tests, 4.5.2.1 through 4.5.2.8.

(3) Test results of initial and final quality conformance tests.

Table IV

Preproduction and Periodic Production
Test Schedule

	Sample and Test Sequence		
	Sample No. 1,4, Etc.	Sample No. 2,5, Etc.	Sample No. 3,6, Etc.
Initial Quality Conformance Tests and Examinations (4.7)	1	1	1
Vibration (4.5.2.1)	2*		2*
Acceleration (4.5.2.3)			3*
Shock (4.5.2.2)		2*	4*
Temperature Nonoperating (4.5.2.4.1)		3	5

*Indicates visual examination after test,

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

Table IV (Cont'd)

Preproduction and Periodic Production
Test Schedule

Test	Sample and Test Sequence		
	Sample No. 1,4, Etc.	Sample No. 2,5, Etc.	Sample No. 3,6, Etc.
High Temperature Operating (4.5.2.4.2)	3		6
Humidity (4.5.2.5)		4	7
Pressure (4.5.2.8)	4		8
Enclosure (4.5.2.6)	5		9
Low Temperature Operating (4.5.2.4.3)		5	10
Life Test (4.5.2.7)			11
Final Quality Conformance Tests and Examinations (4.7)	6	6	12

4.5.1 Preproduction test equipment. The following items of test equipment are required to perform preproduction tests specified herein.

4.5.1.1 A vibratiion table capable of vibrating the motor and necessary test fixtures at 10 g's maximum over a frequency range of 5 to 500 Hertz with a maximum double amplitude of 0.2 inch and with provisions for controlling environmental temperature in the range of -45°F to +14°F.

4.5.1.2 A shock testing machine capable of applying shocks of up to 300 g's maximum at durations from 0.5 to 20.0 milliseconds.

4.5.1.3 A centrifuge capable of exerting acceleration forces up to 35 g's magnitude for periods of 1 minute.

4.5.1.4 A temperature chamber capable of programmed temperature variations over a range of -40 to ±200 degrees F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

4.5.1.5 A humidity chamber capable of providing a constant humidity condition of 95 percent at +110 degrees F for periods up to 8 hours.

4.5.1.6 Motor life testing equipment, including regulated power supplies, switching circuitry, and loading fixtures, to operate the motor through the duty cycle as specified in 3.3.3. The setup should be adaptable to tests of both 4.5.2.4.2 and 4.5.2.7.

4.5.1.7 A vacuum and pressure chamber capable of operation from 2.7 to 20.0 psia with provision for traversing this range from the lower to the higher limits at the rate of 0.5 psi per second.

4.5.1.8 Items of test equipment specified in 4.7.

4.5.2 Detailed preproduction tests

4.5.2.1 Vibration

4.5.2.1.1 Nonoperating. The packed motor (see 5.2) shall be placed on a vibration platform that is fenced on all sides sc that the pack cannot tip over or fall off the platform. With the packed motor in the vertical position, gradually increase The frequency from 5 Hz until the pack just begins to leave the platform. At this point, the frequency shall be maintained and the tests will be continued for 1 hour.

4.5.2.1.2 Ready nonoperating and ready. The unpackaged motor, mounted on a support frame, shall be rigidly attached to the vibration equipment. This support frame shall be capable of transmitting the vibrations listed in Table I to the motor along the axes shown in Figure 1. Vibration shall be conducted at three temperature conditions: the motor stabilized at room temperature, at -45 ±5°F and at +140 ±5°F. The resonant modes for each of the three temperature conditions in each of the three axes shall be established by varying the frequency of applied vibration slowly from 5 to 500 Hz in accordance with Table I, for each temperature condition and in each axis. If more than four resonances are encountered for any one temperature condition on any one axis, the four most severe resonance points shall be chosen for the tests at resonance, for that one axis and that one temperature condition. The total vibration time in each axis shall be 180 minutes, with the motor for 60 minutes of this time stabilized at -45 ±5°F, 60 minutes stabilized at +140 ±5°F, and 60 minutes at room temperature. The motor shall be vibrated for 10 minutes at each resonance point for each of the specified temperatures in each axis; the remainder of the 60 minutes shall be used in cycling per Table V. The frequency rate during cycling shall be logarithmic such that 1/12 of the total vibration time is required to proceed from the lower extremity of frequency range.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

Table V

Resonant and Cycling Schedule

No. of Resonances	Time at resonance at each temperature in each axis, and at each resonance (Min)	Cycling time at each temperature in each axis (Minutes)
0	0	60
1	10	50
2	10	40
3	10	30
4	10	20

4.5.2.2 Shock

4.5.2.2.1 Nonoperating. The packed motor (see 5.2) shall be subjected to free fall drop tests on corners and surfaces of the container. The packaged unit shall be dropped from a height of 36 inches above an impact surface which shall be essentially nonyielding stone, concrete, or similar material. The container, with the packaged unit inside, shall be dropped in sequence on corners 5-1-2, 6-3-4, 5-2-3, and 6-1-4, and side 1 and face 5 as defined in Figure 2.

4.5.2.2.2 Ready nonoperating and ready. Mount the unpackaged motor on to a shock test fixture (Figure 1) so that a minimum of 0 percent of table input of the fixture will be applied to the motor under test. Subject the motor to shocks listed in Table II, in the conditions specified. Duration shall be measured by an accelerometer mounted on input table of fixture to indicate applied shock.

4.5.2.3 Acceleration. The motor shall be rigid mounted and subjected to acceleration magnitudes and directions required by Table III for 1 minute at room temperature for each condition specified.

4.5.2.4 Temperature

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

4.5.2.4.1 Temperature, nonoperating and ready nonoperating. The unpackaged motor shall be placed in a temperature controlled chamber. The chamber shall then be subjected to the following temperature cycles:

(a) Heat chamber to +140 +5 -O degrees F in 2.5 ±0.5 hours.

(b) Maintain at +140 +5 -O degrees F for 4 hours minimum, then within 2.5 ±O.5 hours cool chamber to -45 +0 -5 degree F.

(c) Maintain at -45 +0 -5 degrees F for 4 hours minimum,then within 2.5 ±0.5 hours return chamber to room temperature.

Repeat this schedule for a minimum of three cycles.

4.5.2.4.2 High temperature operating. The motor shall be operated for one or more duty cycles as specified in 3.3.3 (load included) at an ambient temperature of 200 ±5 degrees F. The motor shall be conditioned in the test chamber at 200 ±5 degrees F for 1 hour prior to the test. After the 10-minute operating period, the motor shall be left in the test chamber and the insulation resistance shall be measured. The resistance reading shall be taken at 7 ± 1/2 minutes following termination of the operating period, using the method of 4.7.3.1, and shall be not less than 100 megohms.

4.5.2.4.3 Low-temperature operating. The motor shall be conditioned in the test chamber at a temperature of O ±5 degrees F for 1 hour prior to testing. The tests to be made in this environment are as follows:

(a) Using a motor operating circuit equivalent to that shown in Figure 3, energize the motor in each direction with the supply voltage adjusted to 28.0 ±O.5 volts and with zero load on the shaft. The steady-state input current shall be determined within 3 seconds after start in each direction and shall not exceed O.5 ampere.

(b) The motor time constant shall be determined as in Section 4.7.3.8. The same criteria shall be met.

4.5.2.5 Humidity. The motor, ready for installation, shall be exposed to an atmosphere of 95 percent relative humidity at a controlled temperature of 110 ±5 degrees F for a period of 8 hours. The motor, at room temperature, shall be placed in the test chamber with the shaft horizontal or inclined downward. After the exposure, the motor shall be wiped dry and placed in room environment for not more than 1 hour and shall then be tested according to 4.7.3.1; the insulation resistance shall exceed 20 megohms. After standing in room environment for not more than 24 hours the insulation resistance shall exceed 100 megohms.

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:07Z
Check the source to verify that this is the current version before use.

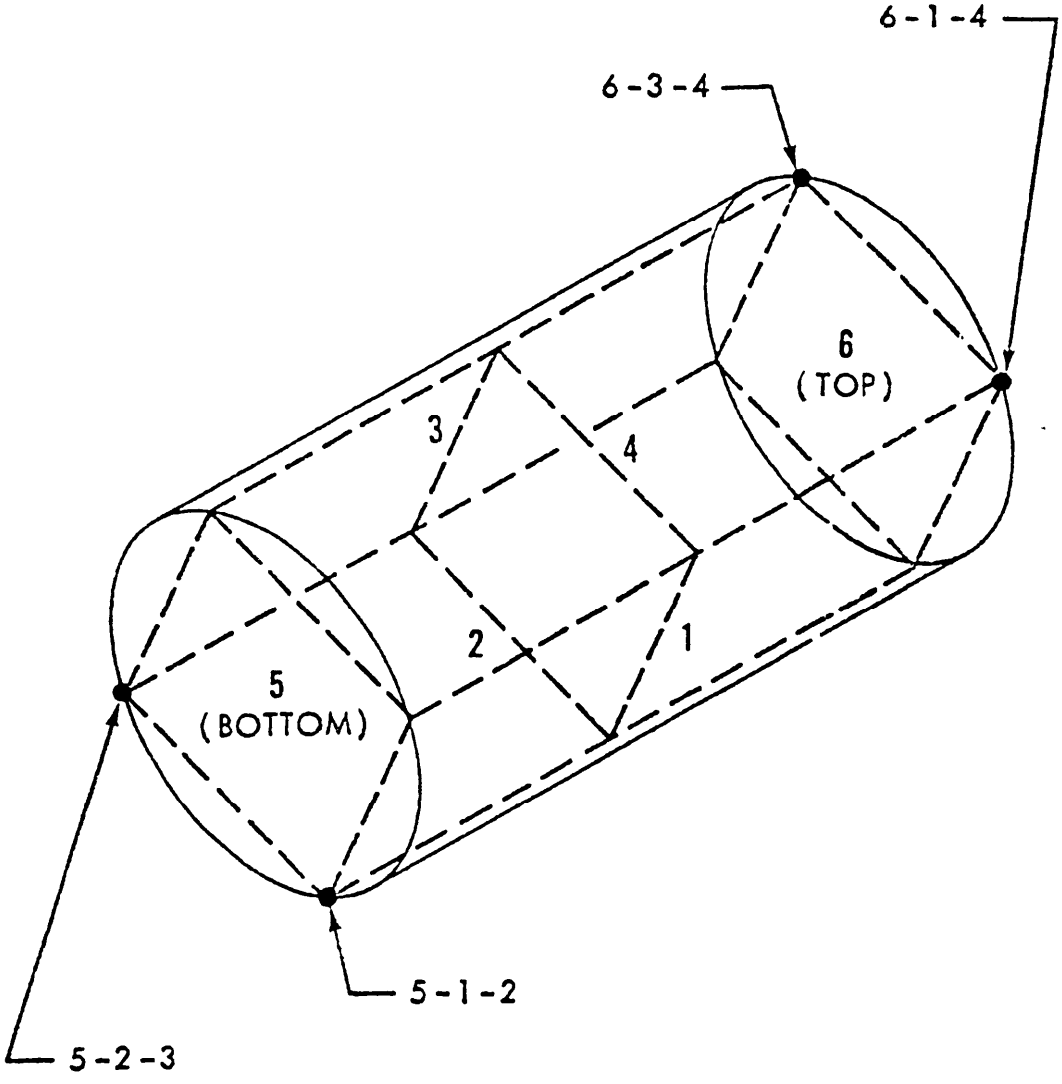


Figure 2. Definition of Faces and Corners of Shipping Container

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

4.5.2.6 Enclosure. The purpose of this test is to verify that the case around the body of the motor is effectively splash-proof. The output shaft of the motor, its bearing, and the adjacent portion of the housing shall be suitably enclosed to prevent entry of moisture. The test shall be similar to the Rain Tests, Procedure 11, as described in MIL-E-5272, with exceptions and/or special provisions as follows:

(a) The water introduced as rain shall be 5 percent sodium-chloride solution prepared as in Federal Test Method Standard No. 151, Method 811, except that tap water may be used instead of distilled water. Alternatively, natural sea water may be used if available and clean.

(b) The rainfall rate shall be reduced to 2.0 ±O.5 inches per hour and the droplet diameters shall be mainly within the range of 2 to 5 millimeters.

(c) variation of direction of rainfall from the vertical is not required but is permitted.

(d) The motor shall be placed with its longitudinal and major axes horizontal, i.e., X and Z axes as shown in Figure 1.

(3) The motor shall be exposed to the rainfall for 10 minutes in each of two positions, 180 degrees apart, for a total test duration of 20 minutes.

After the exposure, the residual salt water shall be washed or flushed off and the motor wiped dry. After standing in room environment for not more than 1 hour, the motor shall be tested according to 4.7.3.1; the insulation resistance shall be not less than 20 megohms. After standing in room environment for a total of 24 hours, the insulation Resistance shall exceed 100 megohms.

4.5.2.7 Life test. The motor shall be subjected to an operating life test consisting of 150 duty cycles as described in 3.3.3.

4.5.2.8 Pressure. The motor shall be placed in the test chamber and the pressure reduced to 2.7 ±O.5 psia, The rate of pressure reduction shall not exceed 0.5 psi per second. The pressure shall then be increased uniformly to 20 ±O.5 psia at the rate of 0.5 ±0.05 psi per second. This pressure cycle shall be repeated five times. The pressure shall be reduced gradually to ambient after the final cycle. The motor housing shall show no deformation from this exposure nor shall the effectiveness of splash-proofing elements be compromised.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

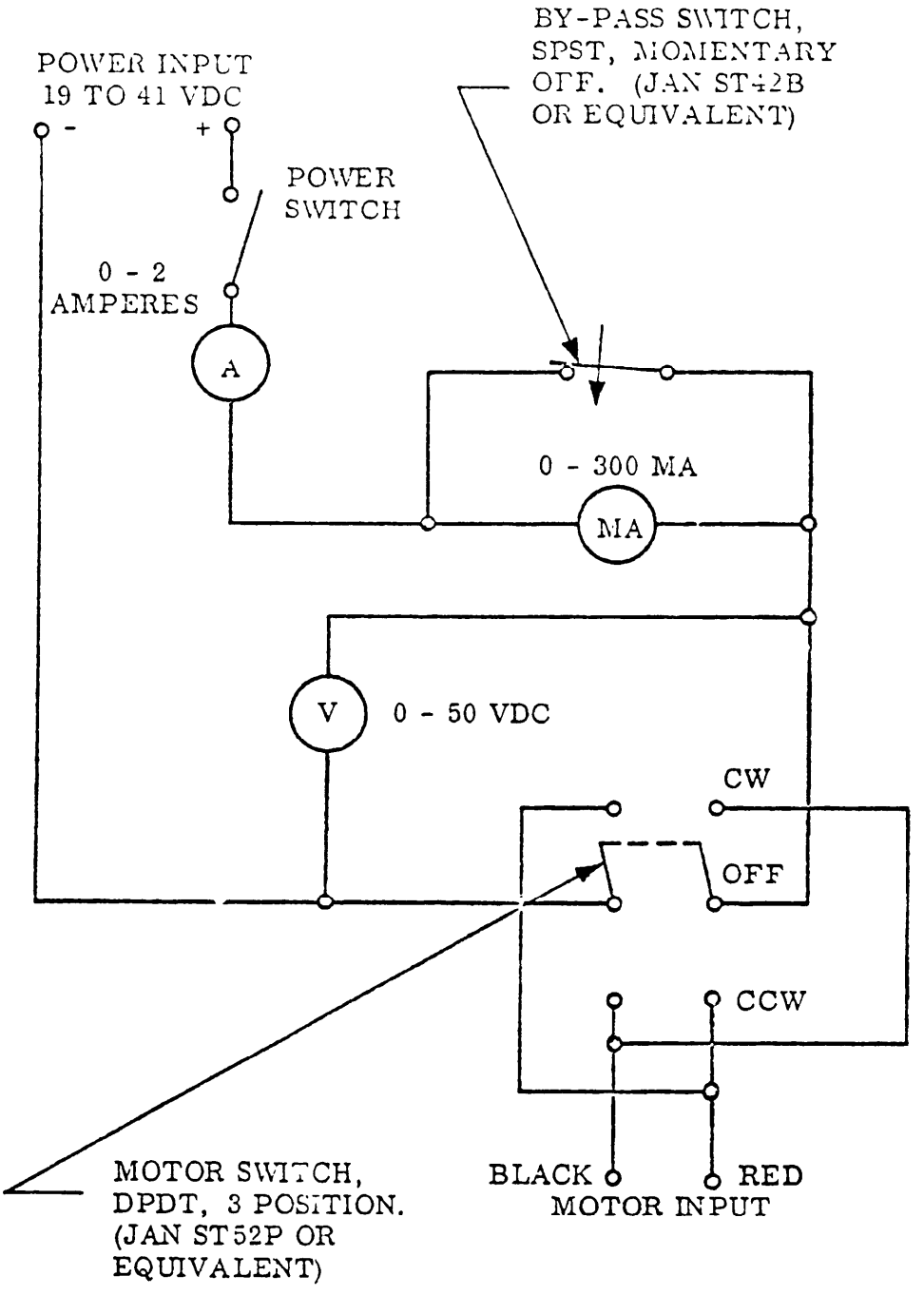


Figure 3. Motor Operating Circuit

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:07Z
Check the source to verify that this is the current version before use.

4.5.3 Preproduction acceptance criteria. Acceptance shall be based on meeting all specified requirements in the preproduction tests and upon satisfactory completion of the final quality conformance tests and examinations. For motors subjected to life test, the insulation resistance and no-load speed requirements are modified in accordance with 3.3.4.

4.6 Periodic production tests. Periodic production samples, selected in accordance with 4.3.2, shall be subjected to tests of 4.5.

4.6.1 Periodic production test equipment. The items of test equipment shall be those specified for the conduction of all tests of 4.5 and 4.7.

4.6.2 Periodic production test schedule. The periodic test sample shall be subjected to the tests noted in Table IV in the order specified.

4.6.3 Periodic production acceptance criteria. Acceptance criteria shall be in accordance with 4.3.2. and 4.7.1.3.

4.7 Quality conformance tests and examinations. Unless otherwise specified each motor delivered under contract shall be subjected to the tests and examinations specified herein.

4.7.1 Conditions

4.7.1.1 Test sequence. Unless otherwise noted, sequence of testing procedures presented herein is mandatory and must be performed in order shown.

4.7.1.2 Test results. Where space is provided in the Report Form (Appendix) to indicate value or specific reading obtained, the specific reading shall be recorded if a limit or limits are given in Limits column for the test. If limits are not stated in Limits col= umn, the individual performing test or inspection shall initial in spaces provided in Reading column indicating that requirements were satisfactorily met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

4.7.1.3 Acceptance criteria. Failure of the end product to pass any examination or test of this section will automatically classify the unit as nonconforming. When nonconforming units are corrected by the contractor, the unit shall be reinspected. The reinspection of a nonconforming unit may be limited to the test or examination which defined the nonconformante, or when so directed by the cognizant inspector, a complete re-examination and retest of the unit may be required. Nonconforming units, which have not been corrected, will be considered for acceptance only upon formal application by the contractor to the Naval Ordnance Systems Command via the cognizant Government representative.

4.7.2 Quality conformance test equipment. Items of test equipment listed in Table VI are required to perform quality conformance tests specified herein. All items of test equipment shall be of sufficient accuracy and precision to permit valid performance of required tests. Test instruments shall be calibrated at regular intervals. Procedures for and results of all calibrations shall be subject to approval of the cognizant Government Inspector. Items having equivalent characteristics may be substituted where necessary. Verify that all test equipment used is identified as being within calibration.

Table VI

Test Equipment

Item	Identification
Dynamic loading device (calibration accuracy of ±2 percent in the range 100 to 300 ounce-inches torque load). Tachometer or rpm pickup (drag load not to exceed 2 ounce-inches and with calibration accuracy of ±1 percent in the range 50 to 150 rpm. Sampling period shall not exceed 3 seconds. Resolution shall be 0.1 rpm or less). DC voltmeter (accuracy ±O.5 volt in range of 38 volts). DC ammeter, (accuracy ±0.05 ampere in range O to 2 amperes). DC milliammeter, (accuracy of ±5 ma in range 200 to 300 milliamperes).	Dynamometer

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:07Z
Check the source to verify that this is the current version before use.

Table VI (Cont'd)

Test Equipment

Item	Identification
Power switching circuit (to operate motor in either direction under manual control. Voltage at motor shall have a step waveform with rise time less than 100 microseconds).	As shown in Figure 4.
DC power supply (regulated, variable voltage, 15 to 42 volts, O to 10 amperes (minimum). The effective source impedance shall not exceed 0.02 ohm).	
Vacuum tube voltmeter (de), O to 1.0 vdc; Input resistance: greater than 50 megohms; Accuracy: ±3 percent.	
Commutation noise test circuit	As shown in Figure 6.
Oscilloscope (precision, dc coupled, with vertical amplifier response to several megacycles and with calibrated sweep rates accurate within ±2 percent. The range 2-5 milliseconds per centimeter is of primary interest. Sweep triggering facilities are necessary.)	Tektronix Model 531 with plug in preamplifier Type A
Megohmmeter (200 vdc potential, reading accuracy within ±5%.)	
Ohmmeter	Simpson Model 269 or equivalent

4.7.3 Test procedure. The following tests shall be performed on each motor to be delivered:

(a) Insulation resistance of windings (4.7.3.1)

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:07Z
Check the source to verify that this is the current version before use.

(b) No-load speed full voltage (4.7.3.2)

(c) No-load current at full voltage (4.7.3.3)

(d) No-load speed at half voltage (4.7.3.4)

(3) Speed versus load at full voltage (4.7.3.5)

(f) Stall current (4.7.3.6)

(g) Motor stability (4.7.3.7)

(h) Acceleration time (4.7.3.8)

(l) Lead pull test (4.7.3.9)

(j) Commutation noise test (4.7.3.10)

4.7.3.1 Insulation resistance of windings. The insulation resistance shall be measured from either motor terminal to the frame, using a 200 vdc megohmmeter. The reading shall be taken at 15 ±5 seconds following application of the megger voltage. The resistance shall be not less than 100 megohms. (see 3.3.4 for exception after life test. It is permissible to use a conventional ohmmeter (Table VI) for resistance measurements of less than 1 megohm.)

4.7.3.2 No-load speed at full voltage. The motor output shaft speeds shall be measured at no-load (less than 2 oz-in) and at 38.0 ±0.5 vdc in both directions. The average of the two speeds shall be 115 to 125 rpm. The speeds in opposite directions shall not differ by more than 4 rpm. The motor rotation shall conform with 3.3.2.9. The motor operating circuit shall be equivalent to Figure 3 for this test and all following operating tests except as noted. At least three readings should be taken in each direction to check constancy of speed and to determine the mean value. This technique should be used whenever speeds are measured under no-load conditions.

4.7.3.3 No-load current at full voltage. With motor operating steadily at full no-load speed (as in 4.7.3.2) the motor currents for both directions shall be measured. The average for the two directions shall be less than 0.220 ampere. The difference of the two currents shall be less than 15 percent of the average value.

4.7.3.4 No-load speed at half voltage. The motor output shaft speeds " shall be measured at no-load and at 19.0 ±0.5 vdc in both directions. The average of the two speeds shall be within 45 to 55 percent of the average full-voltage speed determined under (4.7.3.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

4.7.3.5 Speed versus load at full voltage. The motor output shaft shall be connected to the dynamometer, using a self-aligning coupling, and thence to the tachometer or rpm counter. The motor shall be energized at 38.0 ±0.5 volts, the loading adjusted to 100 ±2 ounce-inches and the resultant speed and current measured, for each rotational direction. These speeds shall be greater than 83 percent of the average no-load speed determined in 4.7.3.2. The same tests shall be performed for loading of 300 ±6 ounce-inches. These speeds shall be greater than 50 percent of the average no-load speed. In order to minimize the effects of armature heating, the motor shall be stabilized to room temperature before each trial. The application of the load and data reading should be completed as expeditiously as possibly and certainly within 10 seconds.

4.7.3.6 Stall current. The motor current for stall condition shall be determined, either graphically or by computation, as the linear extrapolation of the currents measured in 4.7.3.5 at the 100 and 300 ounce-inch loads. The average of currents for the two directions of rotation shall be used. For computation of stall current (I_2) the following formula shall be used:

$$I_s = I_1 + \frac{S_1}{S_1 - S_3}(I_3 - I_1)$$

where

I_1 = Average current at 100 oz-in.

I_3 = Average current at 300 oz-in.

S_1 = Average speed at 100 oz-in.

S_3 = Average speed at 300 oz-in.

The value of stall current, determined as above, shall net exceed 3.3 amperes.

4.7.3.7 Motor stability. The purpose of this test is to prove the stability of the permanent magnet field during repeated instantaneous reversals (“plugging") of the motor at maximum input voltage. The motor shall be instantaneously reversed with an input of 41.0 ±0.5 volts until at least 20 reversals have been made. The motor shaft shall be uncoupled for this test. The motor shall accelaerate and run for at least one second between successive reversals. The no-load speed shall be determined prior to and after this test, using the method of 4.7.3.2. The "before" and "after" speeds shall be equal within the accuracy of measurement. If both “after” speeds exceed the “before” speeds by 1 rpm or more, the motor stability test shall be repeated, but only once. The final speeds must meet the criteria of 4.7.3.2.

4.7.3.8 Motor time constant

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

(a) Connect input A and external trigger of oscilloscope to SCOPE INPUT terminals of Figure 4. Position oscilloscope control as follows:

Control	Position
INPUT SELECTOR	A, DC
TRIGGER MODE	AC Fast
TRIGGER SLOPE	EXT+
TRIGGER STABILITY	FULLY CW
TRIGGER LEVEL	FULLY CW
TIME/CM	100 MICRO SEC
5X MULTIPLIER	OFF
MULTIPLIER	5
MILLIVOLTS/CM	10
MV/CM MULTIPLIER	2
VARIABLE	CALIBRATED (FULLY CW)

(b) Accelerate motor from rest with no-load with the motor connected to the circuit shown in Figure 4. A continuous sweep should appear on the oscilloscope. Turn sweep stability control on oscilloscope in a CCW direction until trace just disappears. Place MOTOR REVERSING switch (Figure 4) to CW position and depress MOTOR START switch (Figure 4) to cause motor to run. Continue derressing and releasing MOTOR START switch and adjust triggering level control in a CCW direction until a single sweep is obtained when MOTOR START switch is depressed.

(c) Position TIME/CM switch on oscilloscope to 10 MILLISEC. Depress MOTOR START switch. Using the VERTICAL POSITION, MV/CM MULTIPLIER, and VARIABLE controls on the oscilloscope, position steady state motor current trace at bottom line on graticule and peak motor current trace 5 cm above bottom line on graticule (see Figure 5).

(d) Position TIME/CM switch on oscilloscope to 1 MILLISEC. Position HORIZONTAL POSITION control on oscilloscope so that trace crosses graduated vertical center line between 1.6 and 1.8 cm above bottom line when MOTOR START switch is depressed (see Figure 5). Record distance from triggering point to vertical center line. Calibration is 5 ins/cm. The time constant shall be less than 12 ms. The reading shall be taken with an accuracy of 1.0 ms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

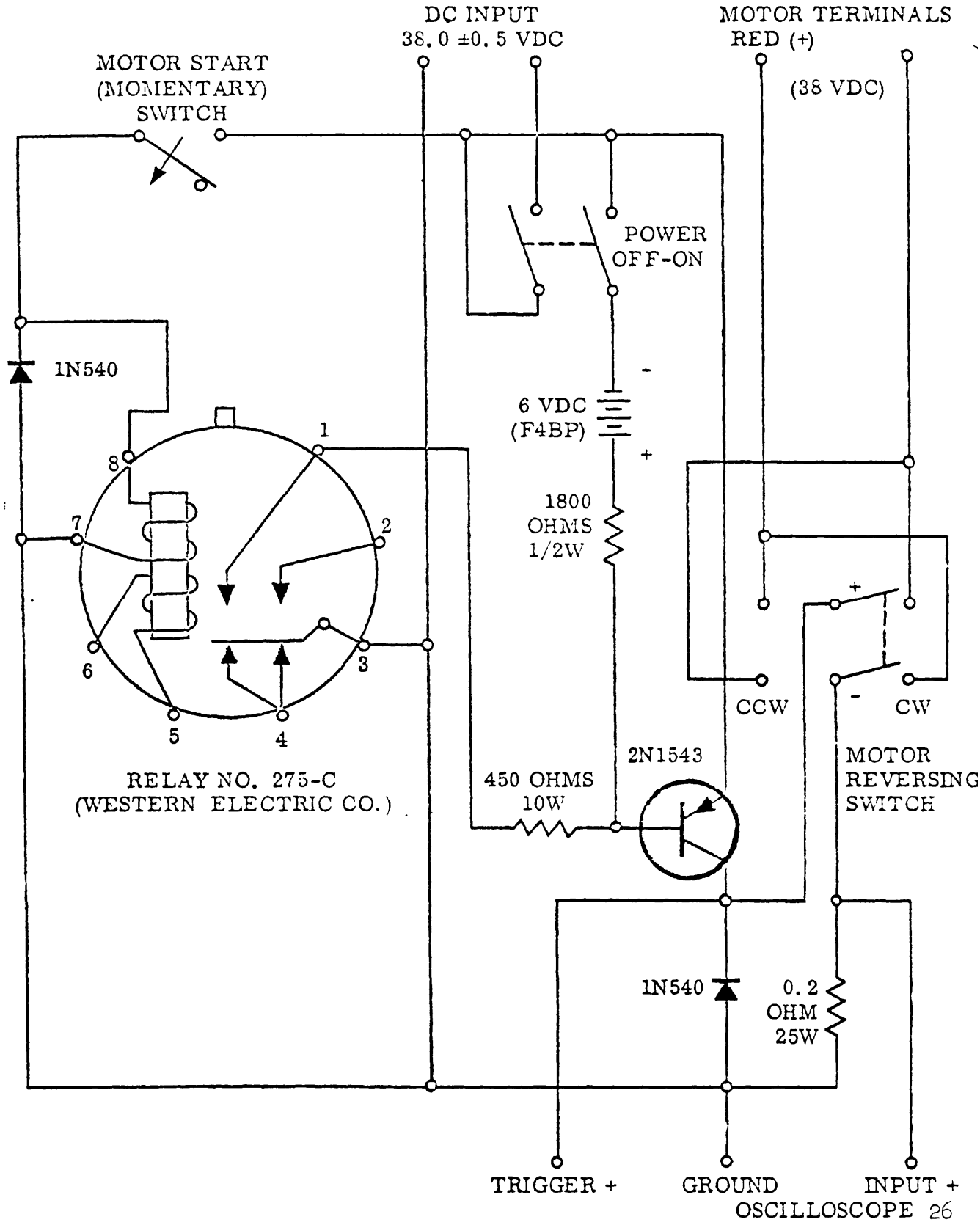


Figure 4. Power Switching Circuit

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

(e) Position TIME/CM switch on oscilloscope to 10 MILLISEC. Place MOTOR REVERSING switch to CCW position. Depress MOTOR START switch. Using VERTICAL POSITION, MV/CM MULTIPLIER, and VARIABLE controls on oscilloscope, position steady state motor current trace at bottom line on graticule and peak motor current trace 5 cm above bottom line on graticule (see Figure 5).

(f) Position TIME/CM switch on oscilloscope to 1 MILLISEC. Position HORIZONTAL POSITION control on oscilloscope so that trace crosses graduated vertical center line between 1.6 and 1.8 cm above bottom line when MOTOR START switch is depressed (see Figure 5). Record distance from triggering point to vertical center line. (Calibration is 5 ms/cm.) The time constant shall be less than 12 ms. The reading shall be taken with an accuracy of 1.0 ms.

4.7.3.9 Terminal stress test. Each motor terminal shall be pulled by attaching a spring scale at the free end. The terminal shall be pulled in one or more directions perpendicular to its longitudinal axis. Each terminal shall withstand a force of 5 pounds without breaking or loosening.

4.7.3.10 Commutation noise. Using the filter-rectifier circuit and equipment setup as diagramed in Figure 6, measure the noise level signal as follows:

(a) Set power supply voltage at 38.0 ±1.0 volts.

(b) Start motor in clockwise direction. Observe noise level signal on VTVM for 30 seconds; determine that maximum indication which is sustained for at least one second. Record the voltage indication for CW direction.

(c) Start motor in counterclockwise direction. Make same observation as in (b). Record the voltage for CCW direction.

The noise level voltage shall not exceed 0.75 volt for either direction of rotation.

4.7.4 Visual inspections

4.7.4.1 Identification marking. The nomenclature, control callout and identification data shall be correct, clear, and legible and in accordante with the applicable specification control drawing.

4.7.4.2 Examination of product. Each motor shall be examined for conformante with 3.5 and with the specification control drawing listed in 2.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

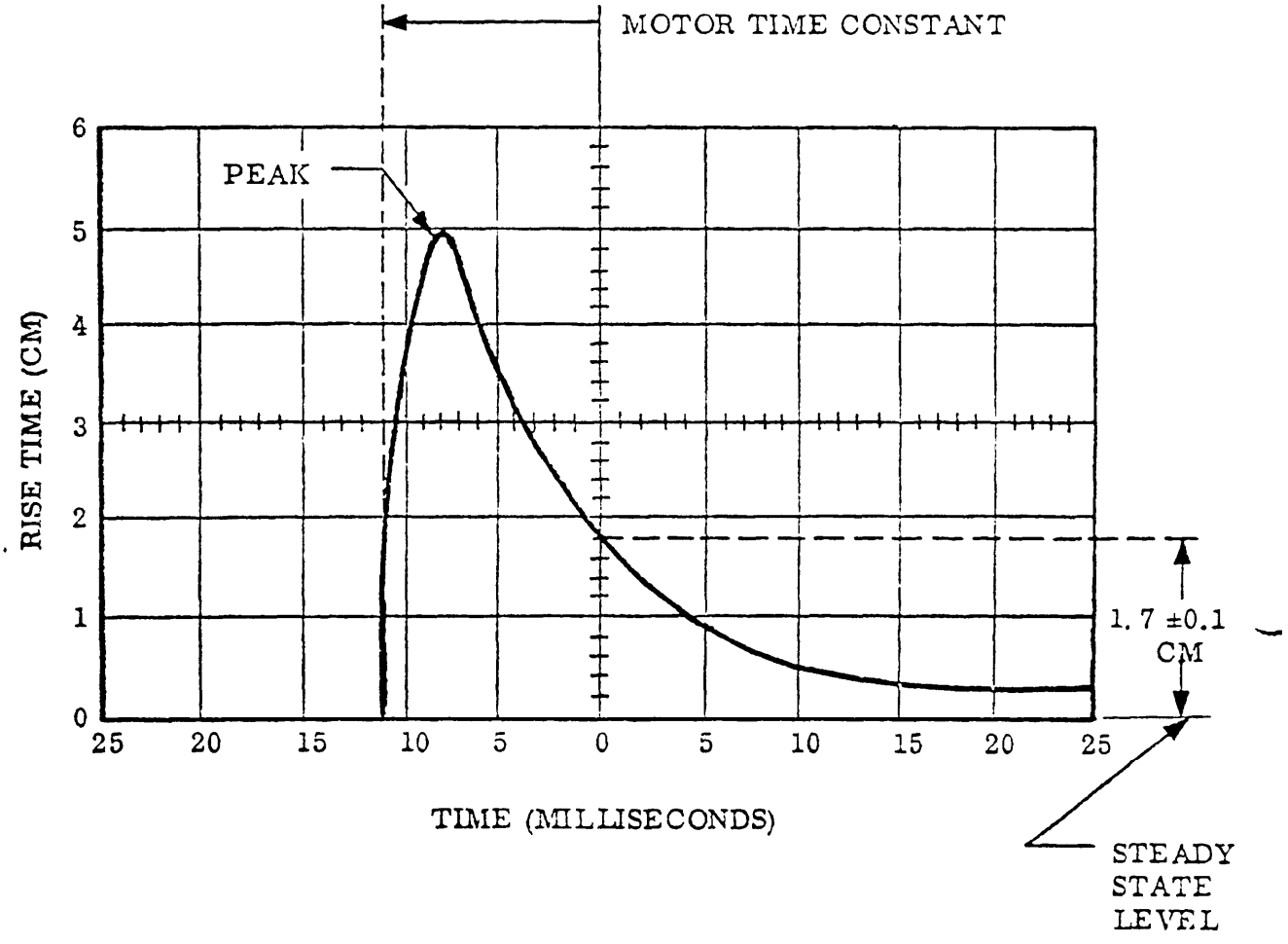


Figure 5. Motor Time Constant

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

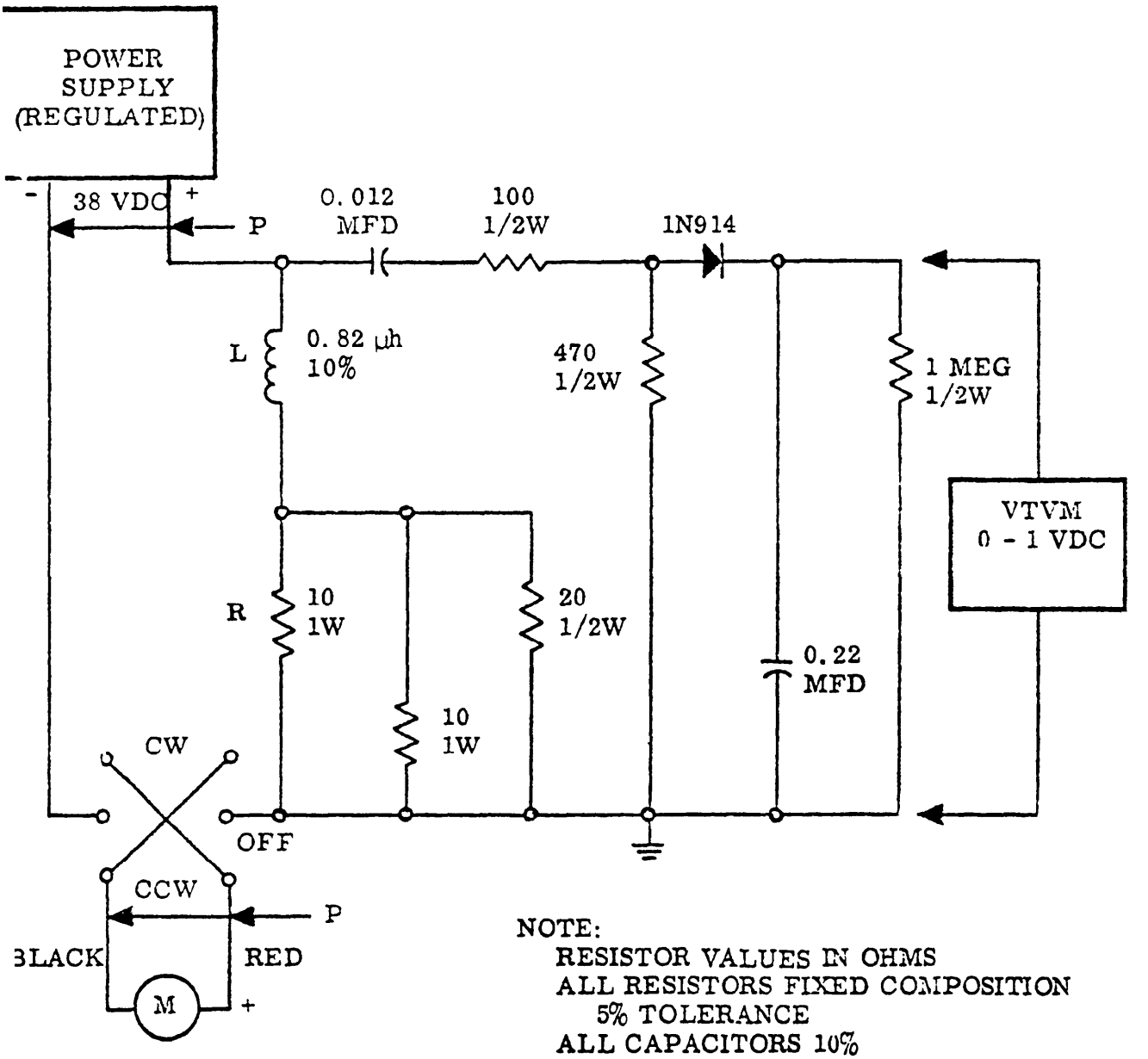


Figure 6. Commutation Noise Test Setup

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:07Z
Check the source to verify that this is the currentversion before use.

4.7.5 Cleanliness. The presence of foreign material in the assembly shall classify the unit as defective. Foreign material includes anything not intended to be present.

5. PREPARATION FOR DELIVERY

5.1 Reservation and packaging. Reservation and packaging shall be Level A or C, as specified (see 6.2.)

5.1.1 Level A. Cleaning, drying, preservation, and packaging methods for Level A shipment shall be in accordance with the instructions contained in MIL-P-116, process C-1 and Method IIe.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation, and packaging shall be sufficient to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity and may conform to the supplier's commercial practice.

5.2 Packing

5.2.1 Level A. The motor, preserved and packaged in accordance with 5.1.1, shall be packed in quantities of nine, in a container in accordance with PPP-B-636, Type CF, Class Weather Resistant.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The motors, packaged as specified in 5.1.3, shall be packed in a reamer which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Containers and packing shall comply with uniform freight classification rules, or with rules and regulations of other carriers as applicable to the mode of transportation.

5.3 Marking

5.3.1 Normal marking. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use and description

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:07Z
Check the source to verify that this is the current version before use.

6.1.1 Intended use. The motor is to be used to actuate the control surfaces in an automatically controlled steering system.

6.1.2 Description. The motor is a direct current type with integral gear reduction.

6.1.3 Storage life. The motor when packaged and preserved for overseas shipment shall be capable of meeting all requirements after a minimum period of 5 years.

6.1.4 Installed life. The motor, in the ready, nonoperating condition, shall be capable of meeting all requirements after a minimum period of 1 year.

6.1.5 Design considerations. While stall torque is not a requirement of this specification, it is a useful design parameter which may be employed, with no-load speed, to summarize the speed/torque capabilities of the motor. Assuming a linear relationship of speed and torque, the stall torque obtained by extrapolation horn speed versus load data (4.7.-3.5) is intended to be at least 600 oz-in.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Required level of preservation and packaging.

(c) Required level of packing.

(d) Assigned activity for preproduction and periodic tests.

(e) Required data.

(f) Any exceptions to this specification.

(g) All additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any sample(s) to pass the prescribed inspections.

6.3 Definitions. For the purpose of this specification, the following definitions apply:

6.3.1 Vibration. Vibration is a sinusiodal motion defined by amplitude and frequency. The amplitude may refer to displacement or acceleration. The displacement is given in inches peak-to-peak or the corresponding maximum acceleration in gravity units (g). The frequency is given in Hertz (Hz).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

6.3.2 Motor condition

6.3.2.1 Nonoperating. The condition of the uninstalled motor when stored, transported, or handled (see 5.2).

6.3.2.2 Ready, nonoperating. The condition representative of the motor when ready for assembly or assembled in a torpedo when being issued, transported, or handled.

6.3.2.3 Ready. The condition representative of the motor when fully assembled with the end item and installed on the lauching mechanism. prior to operating.

6.3.2.4 Operating. Operating is the condition of the motor when operating.

6.3.3 Shock. Shock is an abrupt change "in the motion of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse in units of gravitational acceleration (g's), duration in units of time, milliseconds ms , and a shock wave approximating a one-half sine wave. The time duration shall be measured between two points where two lines intersect the zero g line. These lines are tangent to the shock wave slopes.

6.3.4 Acceleration (steady state) .Acceleration (steady state) is the the rate of change of velocity of a body resulting from the application of a sustained external force with a uniform buildup to peak in more than 0.010 second. Acceleration is given in gravitational units (g).

6.3.5 Temperature. Temperature of the environment to chich the motor is exposed as expressed in degrees Fahrenheit (^{0}F),

6.3.6 Pressure. Unless otherwise specified, pressure is the pressure of the environment to which the motor is exposed, expressed in pounds per-square-inch absolute (psia) or pounds per-square-inch gage (psig).

6.3.7 Humidity The relative humidity of the environment to which the motor is exposed, expressed as a percent.

6.3.8 Defective unit. A unit shall be considered defective if it does not meet all examinations or tests specified, or deviations are found through visual examination after environmental exposure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TESTS AND EXAMINATIONS

FOR

MOTOR, DIRECT CURRENT

2540053

(For Torpedo Mk 46 Mod 1)

EQUIPMENT STATUS			
CONTRACTOR CERTIFICATION		GOVERNMENT ACCEPTANCE	
Contractor		Date of Acceptance	
Contract		Accepted by	
Equipment Ser. No.		Title	
Insp. Req. Satisfied		Agency	

Source: https://assist.dla.mil -- Downloaded: 2026-04-23T07:07Z
Check the source to verify that this is the current version before use.

REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TEST AND EXAMINATION

MOTOR, DIRECT CURRENT

10. SCOPE

10.1 This Report Form contains data sheets to record the results of the quality conformance tests and examinations of 4.7 of this specification. When completed and the authorized certification and acceptance signatures are affixed, this report form or copies thereof shall accompany the motor and shall include the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test sequence. The quality conformance tests and examinations of 4.7 shall be completed and the results recorded or verified in this report form prior to acceptance of the motor. Unless otherwise noted the sequence of testing, recording, and verification, as performed in 4.7, is mandatory and must be performed in the order shown.

20.2 Recording of results. Where space is provided to indicate the value of specific reading obtained, the specific reading will be recorded if a limit or limits is given in the Limit column for the test. The initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated in the Limit column, the individual performing the test or inspection shall initial in the space provided in the Reading column indicating that the requirements have been met satisfactorily.

30. QUALITY CONFORMANCE TEST AND EXAMINATION

	Reference	Limits	Reading
30.1 Electrical tests.			
30.1.1 Test equipment.	4.7.2	In calibration	_____
30.1.2 Test procedure.	4.7.3		
30.1.3 Insulation resistance.	4.7.3.1	Greater than 100 megohms	_____

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

30.1.4 (Continued)		Reference	Limits	Reading
	No load speed, 38 vdc input.	4.7.3.2		CW_____
				CCW_____
			Average 115 to 125 rpm	AVG_____
	Speed difference, opposite directions		4 rpm max	Diff_____
	Rotation vs polarity	4.7.3.3	CCW as marked	_____
30.1.5	No load current.			CW_____
				CCW_____
			Average 0.220 amp max	Avg_____
	Current difference, opposite directions		Less than 15% of average value	Diff_____
30.1.6	Speed, 19 vdc input.			
		4.7.3.4		CW_____
				CCW_____
			Average within 45 to 55% of 4.7.3.2 average	Avg_____
30.1.7	Speed versus load (at full voltage).			

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

Torque in ounce- inches	Speed in rpm		Limits, text paragraph	Current in amperes (Record indication)	
	CW	CCW		CW	CCW
No load			4.7.3.2		
100			4.7.3.5		
300			4.7.3.5		

		Reference	Limits	Reading
30.1.8	stall current.	4.7.3.6	3.3 amperes	_____
30.1.9	Motor stability.	4.7.3.7		
	Speed at 38 volts input before test (20 reversals)			CW____
				CCW____
	Speed at 38 volts input after test		Less than 1 rpm increase from before test readings	CW____
				CCW____
	Motor Stability test-repeat (conditional--see test)			
	Speed before	4.7.3.7		CW____
				CCW____
	Speed after test		Less than 1 rpm increase from before test readings	CW____
				CCW____
			Average of final speeds within 115 to 125 rpm	Avg____

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

	Reference	Limits	Reading
30.1.10 Motor time constant.	407.3.8	Less than 12 milliseconds	
	(d)		CW____
	(f)		CCW____
30.1.11 Terminal stress.	4.7.3.9	No loosening or damage at 5 lb force	_____
30.1.12 Commutation noise.	4.7.3.10	0.75 volt	CW____
			CCW____
30.2 Visual inspections.	4.7.4		
30.2.1 Identification marking.	4.7.4.1	Per drawing	_____
30.2.2 lamination Of Product.	4.7.4.2	Per specification and paragraph 3.5	_____

Preparing Actitity
Navy - OS
Project No. 1355-N289

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-M-82433A(OS) Motor, Direct Current

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:07Z
Check the source to verify that this is the current version before use.

FOLD

7

POSTAGE AND FEES PAID

Naval Ordnance Station
Indian Head, Md. 20640

OFFICIAL BUSINESS

Commanding Officer
Naval Ordnance Station
Standardization Division (QA3)
Indian Head, Maryland 20640

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:07Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:07Z
Check the source to verify that this is the current version before use.